Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [12ME55/13ME55]

FIFTH SEMESTER B.E/B.TECH. EXAMINATION, DECEMBER-2015

DESIGN OF MACHINE ELEMENTS-II

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Use of Machine Design data handbook is permitted.
3. Missing data may be assumed suitably

1. A closed ring is made of 40mm diameter rod bent to a mean radius of 85mm. If the pull along the diameter is 10,000N, Determine the stress induced on the section of the ring along which it is divided into two parts by the direction of pull and it is shown in fig. (1) **14**

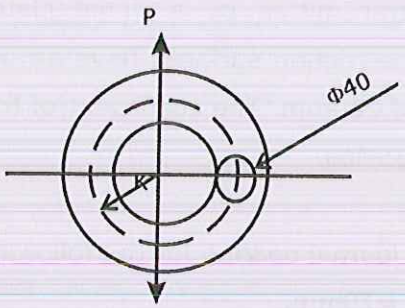

Fig-1

2. **a.** Briefly explain the shearstress induce in a helical compression spring with usual natations. **04**

b. An automobile helical coil spring is to have a mean diameter of 80mm and stiffness of 200N/mm. The total axial force is 8000N and allowable shear stress of spring material is 320N/mm^2, G=80×10^{4}N/mm^2. Determine the following (i) Dia of coil. (ii) Number of active coils. (iii) free length of spring. **10**

3. Select type of V-belt and number of belts required for 10kW, 750rpm induction motor to drive an exhaust fan in a steel plant at 250rpm. the min center distance between shafts is 1.2m. Pitch diameter of motor pulley is 200mm. Assume 10 to 16 hrs/day heavy duty **14**

4. Design a pair of spur gear to transmit 20kW from a shaft rotating at 1000 rpm. to a parallel shaft which is to rotate at 310 rpm. Assume number of teeth on pinion is 31 and 20^0 full depth tooth form. The material for pinion is C-40 steel untreated and for gear cast steel 0.2%C untreated. Determine module, face width, F_t, F_d, F_N **14**

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5. A pair of level gears with 20^0 pressure angle consists of 20 teeth pinion meshing with 30 gear teeth. The module is 4mm while the face width is 20mm. the surface hardness of both pinion and gear is 400BHN. The pinion rotates at 500 rpm and receives power from an electric motor. The starting torque of the motor is 150% of rated torque. Determine the safe power that can be transmitted considering dynamic load, wear strength and endurance strength. The allowable bending stress may be taken as 240MPa **14**

6. Design a worm gear drive to transmit 40kW at 500rpm of worm. The speed ratio is 25. Material for the gear is phosphor bronze and that of the worm is hardened steel. Determine the efficiency of the drive also. **14**

7. a. Explain briefly the uniform pressure theory and uniform wear theory as applicable to friction clutch and brakes. **04**

   b. Determine the power transmitted by a single plate clutch assuming uniform pressure distribution. The friction surfaces have an outside diameter of 350mm, and an inner diameter of 280mm. The coefficient of friction is 0.25 and maximum allowable pressure is 0.85Mpa. **10**

8. Design a hydrodynamic journal bearing for the following particulars: **14**

   Diameter of the Journal =70mm,

   Speed of the shaft = 250rpm,

   Load on shaft = 7500N.

   Viscosity of oil say bolt of 270 sec at 38^0C specific gravity of 0.905 at 15^0C.

   Determine:

   i. Length of bearing.

   ii. Running clearance.

   iii Min film thickness

   iv. Heat dissipation capacity of bearing.

   v. Whether artificial cooling required or not.